

第一类曲线积分

物质曲线的质量

$$\lim_{\lambda \rightarrow 0} \sum_i f(\xi_i, \eta_i) \Delta S_i = \int_C f(x, y) ds$$

空间曲线 C

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta S_i = \int_C f(x, y, z)$$

若 C 为封闭曲线, 第一类曲线积分为

$$\oint_C f(x, y) ds$$

第一类曲线积分的计算

设一元函数 $x = x(t)$ $y = y(t)$

$$\int_C f(x, y) ds = \lim_{\lambda \rightarrow 0} \sum_i f(\xi_i, \eta_i) \Delta S_i$$

$$= \int_a^b f(\varphi(t), \psi(t)) \sqrt{(\varphi'(t))^2 + (\psi'(t))^2} dt$$

$$\sqrt{(x(t_i) - x(t_{i-1}))^2 + (y(t_i) - y(t_{i-1}))^2}$$

$$= \sqrt{(x'(t_i))^2 + (y'(t_i))^2} \cdot (t_i - t_{i-1})$$

若 $y = \varphi'(x)$ $\int_C f(x, y) ds = \int_a^b f(x, \varphi(x)) \sqrt{1 + \varphi'(x)^2} dx$

若为三元 $\int_C f(x, y, z) ds = \int_a^b f(\varphi(t), \psi(t), w(t)) \sqrt{(\varphi'(t))^2 + (\psi'(t))^2 + (w'(t))^2} dt$

例: 计算 $\int_C x ds$, C 是 $y=x^2$ 点 $(0,0)$ 到点 $(1,1)$ 的弧

$$\int_C x ds = \int_0^1 x \sqrt{1+(2x)^2} dx = \frac{1}{12} (1+4x^2)^{\frac{3}{2}} \Big|_0^1 = \frac{1}{12} (5\sqrt{5}-1)$$

例: 计算 $\int_C (x^2+y^2+z^2) ds$ $x = a \cos t$ $y = a \sin t$ $z = kt$

$$\begin{aligned} \text{原式} &= \int_0^{2\pi} ((a \cos t)^2 + (a \sin t)^2 + (kt)^2) \sqrt{(-a \sin t)^2 + (a \cos t)^2 + k^2} dt \\ &= \int_0^{2\pi} (a^2 + k^2 t^2) \sqrt{a^2 + k^2} dt = \sqrt{a^2 + k^2} \left(a^2 t + \frac{1}{3} k^2 t^3 \right) \Big|_0^{2\pi} \\ &= \frac{2\pi}{3} \sqrt{a^2 + k^2} (3a^2 + 4k^2 \pi^2) \end{aligned}$$

例: 计算 $\int_C \sqrt{x^2+y^2} ds$. C 为圆周 $x^2+y^2=ay$

$$\rho^2 = a \rho \sin \theta \Rightarrow \rho = a \sin \theta$$

$$x = a \sin \theta \cos \theta \quad y = a \sin^2 \theta \quad 0 \leq \theta \leq \pi$$

$$\begin{aligned} \int_C \sqrt{x^2+y^2} ds &= \int_0^\pi \sqrt{a^2 \sin^2 \theta} \cdot \sqrt{a^2 (\cos^2 \theta - \sin^2 \theta)^2 + 4a^2 \sin^2 \theta \cos^2 \theta} d\theta \\ &= \int_0^\pi a^2 \sin \theta d\theta = -a^2 \cos \theta \Big|_0^\pi = 2a^2 \end{aligned}$$

例 求 $x=3t$, $y=3t^2$, $z=2t^3$ 从点 $O(0,0,0)$ 到 $A(3,3,2)$ 的弧长.

$$s = \int_{OA} ds = \int_0^1 \sqrt{3^2 + (6t)^2 + (6t^2)^2} dt = 3 \int_0^1 (1+2t^2) dt = 3(t + \frac{2}{3}t^3) \Big|_0^1 = 5$$